



Effects of integrating peer assessment with electromechanical game-based learning on enhancing AI learning and computational thinking of undergraduates

Mu-Sheng Chen & Ting-Chia Hsu

To cite this article: Mu-Sheng Chen & Ting-Chia Hsu (05 Apr 2026): Effects of integrating peer assessment with electromechanical game-based learning on enhancing AI learning and computational thinking of undergraduates, Interactive Learning Environments, DOI: [10.1080/10494820.2026.2653064](https://doi.org/10.1080/10494820.2026.2653064)

To link to this article: <https://doi.org/10.1080/10494820.2026.2653064>



Published online: 05 Apr 2026.



Submit your article to this journal [↗](#)



Article views: 165



View related articles [↗](#)



View Crossmark data [↗](#)



Effects of integrating peer assessment with electromechanical game-based learning on enhancing AI learning and computational thinking of undergraduates

Mu-Sheng Chen  and Ting-Chia Hsu 

Department of Technology Application and Human Resource Development, National Taiwan Normal University, Taipei, Taiwan

ABSTRACT

Peer assessment and AIoT board games have emerged as effective game-based learning approaches for enhancing artificial intelligence (AI) education and computational thinking (CT). This study employs a four-group quasi-experimental design to examine their impact on undergraduate learning using a personal audio classifier (PAC). The AIoT board game integrates CT and AI concepts through collaborative, hands-on learning, incorporating peer assessment, problem-solving scenarios, and paired programming. Results indicate that anonymous online peer assessment fosters critical thinking and reduces learning anxiety. The AIoT board game enhances collaboration, creativity, and the practical application of AI concepts. Peer assessment further supports iterative improvement of PAC applications for gameplay. The integration of peer assessment within problem-solving tasks significantly improves learning outcomes, lowering barriers to AI education and preparing students for a technology-driven future. This study highlights the potential of combining AI instructional tools with peer assessment to deepen students' understanding of AI and CT.

ARTICLE HISTORY

Received 15 March 2025
Accepted 24 March 2026

KEYWORDS

Peer assessment; artificial intelligence education; computational thinking; game-based learning; AIoT board game

1. Introduction

In the rapidly advancing landscape of modern technology, profound impacts have been exerted on numerous domains (Schmidhuber, 2015). Artificial Intelligence (AI) wields a profound influence on various professional fields, such as medical diagnosis, AI-driven job interviews, and big data analysis, among others. The development of these technologies has significantly altered our societal perceptions and perspectives, necessitating the cultivation of individuals' abilities to operate or comprehend these technologies. One of the most fundamental and pivotal concepts therein is computational thinking (CT). In contemporary society, CT has evolved into an indispensable foundational skill for individuals (Yadav et al., 2014).

Research suggests that utilizing game-based learning can enhance and sustain students' motivation (Alt, 2023) and engagement (Clark et al., 2016). Properly designed educational game content can facilitate deeper student thinking. In recent years, numerous researchers have integrated the concept of board games into education. Gee (2005) indicated that by establishing clear learning objectives, board games can become effective tools for promoting self-directed learning, problem solving, and in-depth learning. Due to their collaborative nature, board games enable students to experience educational content through social interaction, peer engagement, and mutual learning (Wu et al., 2014). Board games, as an educational tool, are also regarded as an important medium for practicing computational thinking (Bayeck, 2023). On the other hand, research on AI education points out that visualizing and gamifying AI course materials can effectively enhance students' learning interest (DeNero & Klein, 2010).

The primary value of peer assessment lies in the close interaction among peers. Chang et al. (2021) showed through meta-analysis that peer assessment can effectively improve students' learning effectiveness. In typical classroom scenarios, the number of students often far exceeds that of teachers, resulting in limited individual feedback from instructors. Peer assessment serves as a solution to this predicament, as research indicates that peer feedback can be more direct and personalized compared to feedback from teachers (Topping, 2009). From another perspective, teacher feedback is often perceived by students

as authoritative and less open to challenge, while peer feedback tends to be more diverse. Furthermore, peer feedback can foster interactive discussions among students, even to the extent of promoting mutual engagement (Cole, 1991; Zhan et al., 2022).

Students with no programming experience who are just beginning to learn CT often exhibit lower self-efficacy in CT. Lower CT self-efficacy is frequently accompanied by heightened learning anxiety, which can negatively affect learning achievement. However, prior studies also suggest that appropriate levels of challenge may enhance motivation and persistence despite increased anxiety (Bandura, 1977; Lin & Hou, 2022). Therefore, this study aimed to enhance students' computational thinking in the four dimensions proposed by the British Broadcasting Corporation (BBC) in 2017, namely decomposition, pattern recognition, abstraction, and algorithms, through the utilization of AIoT board games and peer assessment. The following research questions were formulated:

- (1) Will there be significant differences in learning achievement among students exposed to the four different instructional interventions?
- (2) Will there be significant differences in CT self-efficacy among students exposed to the four different instructional interventions?
- (3) Will there be significant differences in learning anxiety and engagement levels among students exposed to the four different instructional interventions?
- (4) What are the behavioral differences among students undergoing different instructional interventions?

2. Literature review

2.1. Computational thinking (CT)

CT has been recognized as a fundamental twenty-first-century competency that supports logical reasoning, problem solving, and innovation across disciplines (Voogt et al., 2015). Wing (2006) first conceptualized CT as a cognitive process of applying computer-science principles to formulate problems and design solutions that can be executed by humans or machines. Wing (2011) further emphasized abstraction as the core cognitive mechanism that enables the transfer of computational reasoning to diverse real-world contexts.

Barr and Stephenson (2011) subsequently extended CT into a broader set of nine conceptual abilities: data collection, data analysis, data representation, decomposition, abstraction, algorithmic thinking, automation, parallelization, and simulation. These dimensions reflect the multidimensional nature of CT in the problem-solving process. Later, Selby and Woollard (2013) synthesized these conceptualizations into four essential elements: logical thinking processes, abstraction, decomposition, and problem-solving patterns. Their work clarified the interconnections among CT subskills and laid a foundation for pedagogical implementation.

Building on these developments, BBC proposed a more pedagogically practical and measurable framework that defines CT as the ability to give precise instructions to a computer on what to do next. The BBC model operationalizes CT through four key dimensions, namely decomposition, pattern recognition, abstraction, and algorithms, and provides a clear structure for classroom instruction and assessment. Compared with earlier frameworks that emphasized theoretical breadth, the BBC framework is concise, teachable, and directly applicable to computational learning activities. Because of its pedagogical clarity and applicability, this study adopted the BBC framework as the foundation for instructional design. All subsequent course activities and assessment instruments were developed based on this framework to ensure that each stage of learning incorporated the four CT dimensions, enabling students to decompose problems, recognize patterns, abstract core principles, and design algorithms to address authentic learning challenges.

2.2. Board game in education

Board games have been shown to aid students in cultivating critical thinking, problem-solving abilities, situation analysis, inference, organization, planning, and communication skills (Hinebaugh, 2009). The use of board games for developing educational competencies is not a contemporary phenomenon; its origins can be traced back to as early as 2200 BC. To date, numerous schools have successfully integrated board

games into their curricula. Many schools in a certain city in the United States have incorporated games such as Monopoly and Checkers to enhance students' mathematical skills. Oktaviane et al. (2021) developed a competitive board game to help students learn to classify snakes. The results found that competitive board games have a positive impact on students and make learning full of interactions and interesting.

Compared to traditional learning, incorporating educational board games into the curriculum can further aid students in developing logical thinking, critical thinking, and deductive reasoning skills (Hinebaugh, 2009). In the past, many subjects such as mathematics, language, history, and science have been taught using conventional methods. However, in order to enhance learning effectiveness, motivate students, and pique their interest, educators have infused elements of board games into the curriculum. This integration allows students to engage in communication with others through board games, thereby enhancing their interpersonal communication skills (Charlier & De Fraine, 2013). On another note, Cheng (2018) employed educational board games for language learning, revealing that such an approach effectively improved students' learning effectiveness while also reducing their learning anxiety.

2.3. Peer assessment

Peer assessment is a mode of evaluation whereby learners assess one another, typically those with similar levels of learning, and has been successfully applied in primary, middle, and high schools, even among very young students and those with special educational needs or learning disabilities (Ehly & Topping, 1998). Furthermore, a substantial body of research demonstrates that peer assessment enhances learning effectiveness (Cartney, 2010; Cartney & Rouse, 2006; Cassidy, 2006; Clegg & Bryan, 2006; Panadero et al., 2016; Topping et al., 2000). Importantly, both assessors and those being assessed benefit from this process. They can identify their own shortcomings through evaluating their peers' work, and can also recognize others' weaknesses, leading to improvement (Yorke, 2003; Zhan et al., 2023). However, in countries where the primary educational approach involves teacher guidance and does not encourage independent thinking, students may not be receptive to peer assessment (Topping, 2023).

In online peer assessment, Cheng and Tsai (2012) found that higher psychological safety, lower goal value diversity, greater trust in oneself as an assessor, and more positive social interdependence contribute to cultivating deeper learning. An important feature of online peer assessment is its anonymity, which presents both advantages and disadvantages (Li, 2017). Previous research has indicated that anonymous peer assessment seems to offer advantages in terms of students' perceptions of the learning value of peer assessment. It can provide more critical peer feedback and enhance the social effects of self-perception. This is particularly evident in higher education and contexts with fewer peer assessment scenarios (Panadero & Alqassab, 2019).

3. Method

3.1. Participants

The participants in this study were first-year university students. The research was conducted in the "Introduction to Computer Science" course, and the age of the students was between 18 and 19 years old. All participants had no prior experience with AI or programming, ensuring that the intervention was introduced on a common baseline. There were 29 students in the peer assessment and AIoT board game group (experimental group 1, EG1), 28 students in the peer assessment group (experimental group 2, EG2), 31 students in the AIoT board game group (experimental group 3, EG3), and the conventional teaching group (control group, CG) had 28 students. In order to avoid other factors from interfering with the experimental results of this research, the same teacher taught all classes, and the content, time and progress were the same. However, due to the COVID-19 pandemic, EG2 was conducted in an online format, while the other groups participated in face-to-face classes.

3.2. Research tools

In order to investigate whether there was a significant difference in the learning achievement of each group, this study used the CT and AI learning achievement test to assess the learners' learning status. Kuder-

Richardson reliability was used to confirm the internal consistency of the achievement test content. The KR-20 reliability values of the pretest and posttest were 0.584 and 0.546 respectively. Although these values were below the conventional threshold of 0.70, they were considered acceptable for exploratory educational research. In addition, expert content validity was used, and was confirmed after discussion with a researcher with more than two decades of teaching experience.

The CT self-efficacy scale used in this study was developed by Yağcı (2019). This scale consists of four aspects, namely problem solving, cooperative learning, creativity thinking and algorithmic thinking. The Cronbach's alpha value of the original scale was .969. In this study, four questions were selected from each of problem solving, cooperative learning, creative thinking and algorithmic thinking, giving a total of 16 questions. The selection was based on the items with the highest reliability in each dimension. The Cronbach's alpha values for this study's sample were: problem solving = .815, cooperative learning = .707, creativity thinking = .804, and algorithmic thinking = .855, indicating acceptable to good reliability. The scale adopted a 5-point Likert scale for scoring.

The scale used in this study was derived from the Technology Acceptance Scale compiled by Venkatesh (2000). The original questionnaire had 44 questions and a total of eight aspects. Chang (2014) adapted the scale into the Learning Anxiety and Learning Engagement Scale, which is used to assess the learning anxiety, playfulness and enjoyment felt by students during the learning process. As the Technology Acceptance Scale was originally designed as a generalized instrument, its items can be flexibly adapted by modifying the subject terms to fit different domains (Venkatesh, 2000). Following Chang (2014), we rephrased items to align with the learning context (e.g. shifting focus from "system use" to "learning activity experience") while maintaining the three-factor structure, ensuring content relevance and construct validity. Taking the three aspects in Venkatesh's (2000) research report, the internal consistency reliability of the scale was 0.91, 0.88, and 0.90 respectively, with 19 items in total, and was evaluated with a 5-point scale.

The study also examined students' learning behaviors during the learning activities. Due to the influence of COVID-19, Experimental Group 2 (EG2) conducted the instruction online, making it impossible to record students' activity behaviors. Learning behaviors exhibited in both inter-group and intra-group activities were captured for the other groups. Based on the actual actions of students recorded in videos, Table 1 presents a corresponding behavior identification encoding scheme for each category. To enhance transparency, the criteria for behavioral coding were further clarified. Active behaviors referred to initiating actions such as proposing ideas, operating devices, or leading group discussions, while passive behaviors referred to responding or following instructions without initiating. A coding manual was established, and sample cases were used during coder training to improve inter-rater consistency and repeatability. An expert with experience in AIoT board games and peer assessment research was invited to validate the coding scheme to ensure the reliability of behavior analysis in the study. The expert, alongside researchers, examined the applicability of codes and their respective descriptions and definitions. Each 30-second interval was identified with a code, and video coding techniques were employed to investigate key actions. Additionally, a trained coder participated in the coding process alongside the researchers. The kappa value of .84 indicated good internal consistency, and any inconsistent results were resolved through discussion between the two coders to reach a consensus.

3.3. Instruction design

The peer assessment was conducted on the BookRoll learning management platform. This process adopted an anonymous and randomized allocation approach. Only students who had submitted assignments to the

Table 1. Coding scheme.

Active/Passive	Code	Description	Example
Passive	LT	Listen to the teacher	Listen to the teacher
Active	GC	Group communication	Discuss course and activity content with team members/observe team members' operation
Passive	AT	Ask teacher	Ask a teacher or teaching assistant a course question
Active	P	Perform	Operate the app or teaching materials/find information in the provided documents
Passive	N	Do nothing	Chat/watch or query non-course related content/sleep
Active	OT	Operation and testing	Practice task operation/use mobile phone to test the implementation results
Active	EO	Express opinion	Answer teacher's questions/share heard stories, examples or experiences
Active	D	Debug	Find problems/fix bugs in the data set

platform were eligible to participate in peer assessment. Therefore, there would not be any missing or mismatched evaluations. During the peer assessment, each student received an anonymous peer's assignment along with a peer assessment form, as shown in [Table 2](#). Student evaluation was performed using the criteria specified in the peer assessment form to assign scores, and qualitative feedback could also be provided.

The tools used in the course were the MIT App Inventor platform, the PAC platform and AIoT board game. The MIT App Inventor platform uses visual coding blocks to allow users to design iteratively, making it easier for users with no programming experience to develop mobile applications. In addition to using building blocks to lower the learning threshold, the MIT App Inventor platform is a cloud-based development environment where the entire programming process is operated online and the results are stored in the cloud at any time. The MIT App Inventor platform has many useful components, such as sensors, major platform connection services, micro databases, Bluetooth connections, and so on. In response to the trend of technological development, the MIT App Inventor AI course has been added to provide AI courses, teaching materials, and a platform for related use.

The PAC platform was developed by the MIT App Inventor team. It simplifies the process of machine learning training and completes the model training in a few simple steps. Through this process, users can understand the operation of machine learning. By using this platform, users can train their own speech recognition models, and the completed model files can be placed in MIT App Inventor for subsequent applications.

The ultimate goal of the AIoT board game is to develop students' CT. As shown in [Figure 1](#), students must compete for the resources they need and complete tasks in an environment with limited resources. The AIoT board game is turn-based, so the strategy of moving each turn is extremely important.

3.3.1. Process of using the AIoT game-based learning system

The instructional design of the AIoT game-based learning system focuses on developing students' understanding of AI concepts and CT skills. The process comprises three key phases: data recording and training, integration with mobile applications, and group-based learning and gameplay, as shown in [Figure 2](#).

Phase 1: data collection and classifier creation. The process begins with students working collaboratively in groups to create personalized audio classifiers. Each group collects audio samples for specific commands, such as "forward," "left," and "right." Using the system's interface, they record and analyze their voices, transforming the audio inputs into spectrograms that represent the data visually.

The system processes the recorded data, generating confidence levels for each classification. This feedback allows groups to refine their recordings and improve the quality of their classifier. Through this collaborative activity, students gain hands-on experience with data collection and training, which helps them understand the foundational principles of machine learning.

Phase 2: integration with mobile applications. In the second phase, each group integrates their created classifier into a mobile application. This step focuses on cultivating students' CT skills by engaging them in the process of designing and building a functional application. Students learn how their trained classifier can be used to perform specific tasks, such as controlling AI robots via voice commands.

The activity emphasizes CT, as students explore how their classifiers can be implemented within the app. By learning to create mobile applications, students gain practical skills and experience the potential of AI technology in real-world applications.

Table 2. Peer assessment form.

Rubrics	1	2	3	4	5
Is there a specification of the types of labels and whether the number of labels is three or more (including three)?	No indications or explanations provided.	Labels with specified types and a quantity of two are not matched in the recognition results.	Labels with specified types and a quantity of two are correctly matched in the recognition results.	Labels with specified types and a quantity of two are correctly matched in the recognition results.	Labels with specified types and a quantity of three or more are correctly matched in the recognition results.
Model recognition rate.	Below 50%.	31–50%	51–70%	71–90%	Above 91%.
Success rate of model recognition results.	Below 50% success rate.	51–65%	66–80%	81–99%	100%

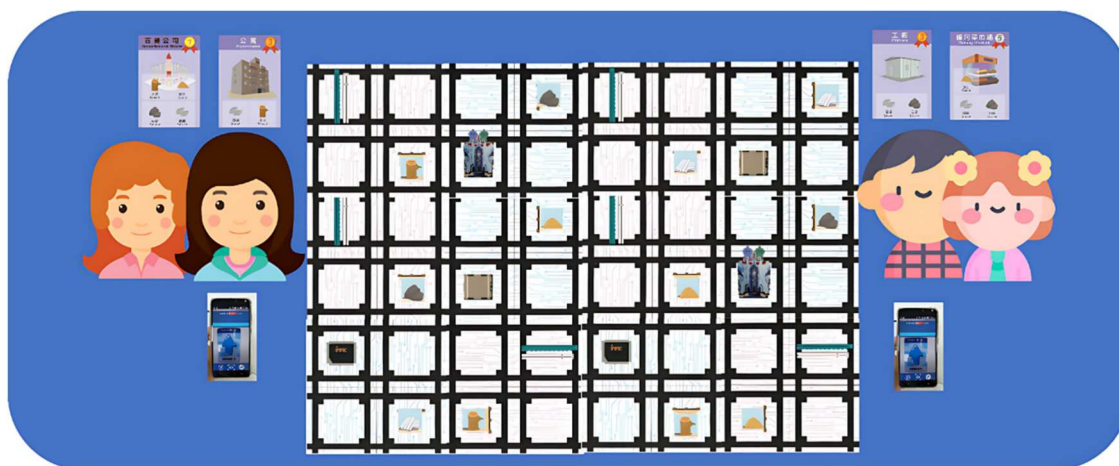


Figure 1. The AIoT board game.

Phase 3: group-based learning and board game gameplay. The final phase of the process extends to group-based collaborative learning. Each group connects their mobile application to the AIoT board game, integrating their classifiers into a tangible, interactive experience. The physical board game includes AI-controlled robots equipped with Bluetooth connectivity, enabling seamless communication between the application and the gameboard.

Groups guide their robots through various challenges on the board, completing tasks that require logical thinking, strategy, and teamwork. This stage not only reinforces AI concepts but also encourages collaboration and problem-solving, as students apply their knowledge to navigate complex scenarios.

3.3.2. Process of the AIoT board game

The AIoT board game is an innovative educational tool designed to cultivate students' CT skills through an engaging, hands-on learning experience. CT, a fundamental skill in problem-solving and critical reasoning,

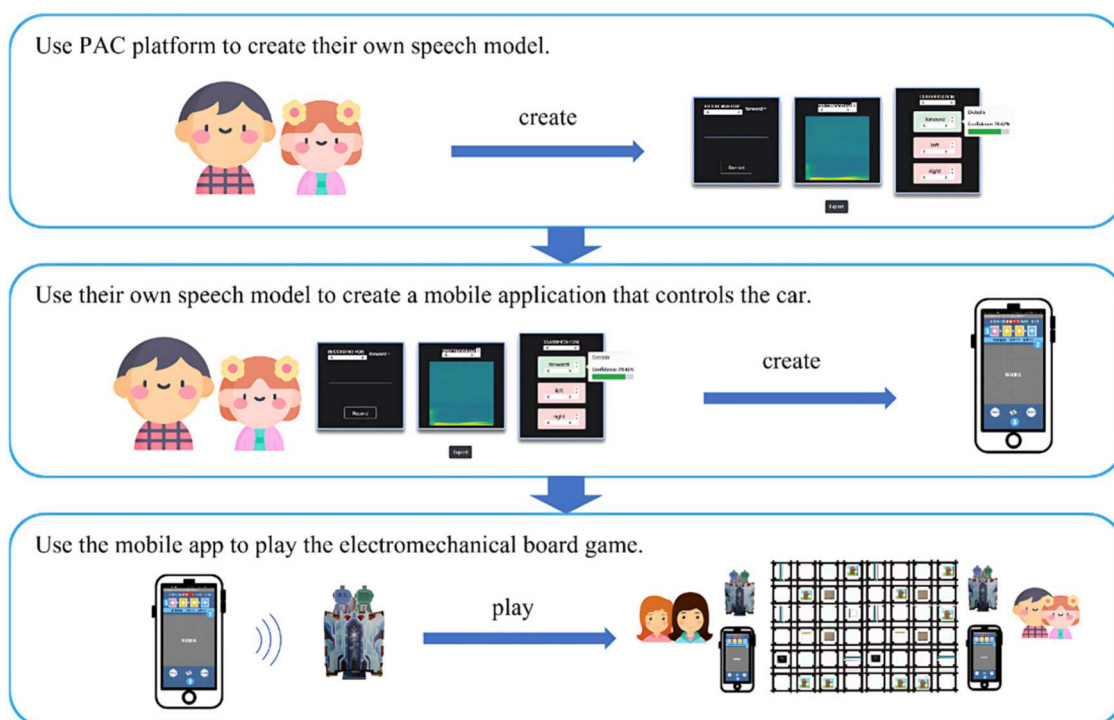


Figure 2. AIoT Game-Based Learning System.

involves breaking down complex problems into manageable parts, identifying patterns, and developing step-by-step strategies to achieve solutions. This board game seamlessly integrates these principles by challenging students to strategize, plan, and execute tasks using smart cars within a dynamic, obstacle-filled environment.

Players are required to analyze the map, anticipate challenges such as obstacles and portals, and devise efficient pathways to collect resources and complete construction tasks. By working collaboratively, students must also communicate effectively, allocate responsibilities, and adapt their strategies in response to evolving situations. These activities mirror the real-world application of CT, such as debugging and optimizing processes, making the learning experience both relevant and practical.

Furthermore, the game incorporates a scoring system that incentivizes higher-level thinking and strategic decision-making. Players learn to weigh the trade-offs between pursuing simpler, lower-scoring tasks or investing in complex, resource-intensive ones for greater rewards. This encourages a deeper understanding of prioritization, resource management, and time optimization. The process of the AIoT board game is shown in Figure 3.

(1) Preparation and Map Arrangement

Players begin by arranging the map tiles. The arrangement is flexible, allowing students to design creative layouts. However, the number of tiles will be limited to fit the playtime. The map includes obstacles and portals to encourage strategic thinking and planning.

(2) Connecting to the Smart Car

Teams use their mobile applications to connect to their smart cars. This step ensures the cars can receive commands from the app, enabling the players to control them during the game.

(3) Setting Up the Smart Car

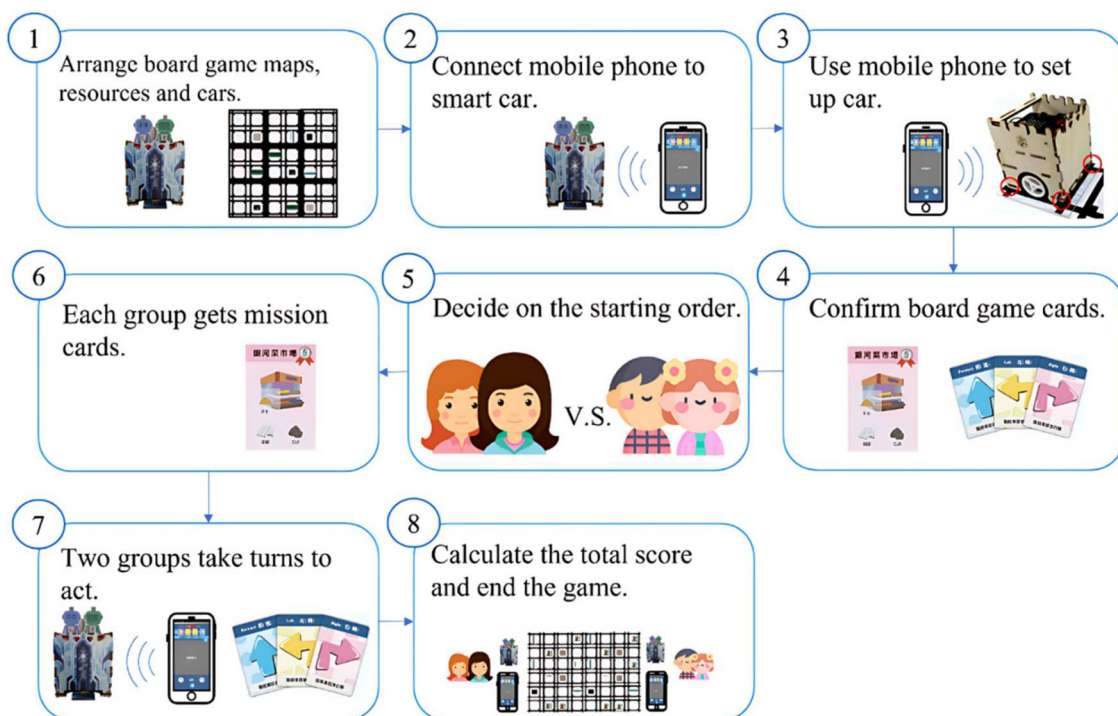


Figure 3. Process of the AIoT board game.

After establishing the connection, players configure their smart cars. This involves calibrating their movement and testing the classifier commands (e.g. "Forward," "Left," "Right") to ensure the car operates correctly. Proper setup is essential for smooth gameplay.

(4) Task Cards and Scoring Configuration

Each team receives task cards outlining construction objectives. Tasks are divided into three levels of complexity, with higher complexity awarding more points: Primary tasks: Require 2 resources, awarding 3 points (average resource value: 1.5 points); Intermediate tasks: Require 3 resources, awarding 5 points (average resource value: 1.67 points); Advanced tasks: Require 4 resources, awarding 7 points (average resource value: 1.75 points). Teams analyze the scoring system to determine the best strategies for maximizing their scores within the time limit.

(5) Determining Play Order

Teams decide the play order, determining who will go first and last. This decision is significant as it affects resource availability and task opportunities, influencing each team's overall strategy.

(6) Team Collaboration and Strategy Planning

Players work together in their teams to plan the robot's route, deciding whether to prioritize quicker, simpler tasks or aim for higher-scoring, complex tasks. They must also consider the dynamic nature of obstacles and portals.

(7) Construction Phase

Once enough resources are collected, teams complete the tasks on their cards. Successfully completing a task adds to their score, and the construction phase reinforces teamwork and problem-solving skills.

(8) Endgame and Scoring

At the end of the game, teams calculate their scores based on tasks completed and resources collected. The team with the highest score wins, encouraging reflection on strategies and gameplay decisions.

3.4. Research procedure

This study employed a quasi-experimental approach as depicted in [Figure 4](#). The study spanned 2 weeks, with each week consisting of 180 min. Prior to the commencement of the first week's sessions, students were grouped and administered a pretest (30 min). The curriculum initiated with an introduction to CT and AI (60 min). Subsequently, machine learning was introduced, and students utilized the PAC platform for constructing and testing speech recognition models (90 min). Grading rules were explained by the instructor, and submission of assignments was required for feedback. Both peer and teacher feedback were provided after the class. EG1 and EG2 utilized a peer assessment form for scoring, while EG3 and the control group (CG) had worksheets for testing their individual models.

During the second week, students began by optimizing their models based on the received feedback (30 min). Subsequently, they started using the MIT App Inventor to create mobile applications. Upon completing their creations, students submitted their assignments and finalized their evaluations or records (60 min). Students then proceeded to optimize their apps based on the received feedback (30 min). In the case of EG1 and EG3, they used the optimized app to play the AIoT board game (30 min), as shown in [Figure 5](#). Conversely, EG2 and CG undertook advanced tasks to modify their mobile apps (30 min), as shown in [Figure 6](#). The week concluded with the administration of the posttest (30 min).

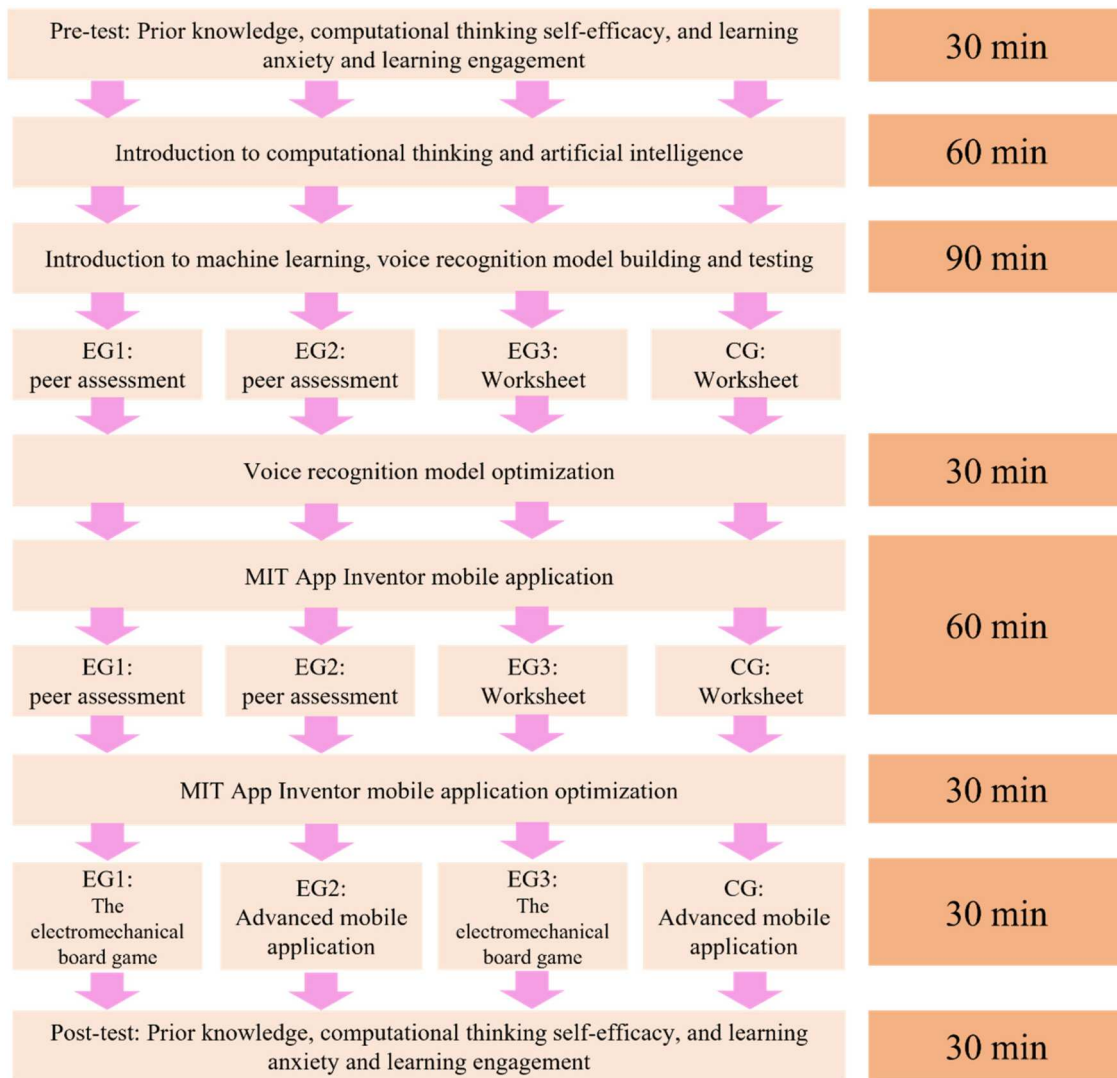


Figure 4. Research procedure.



Figure 5. The AloT board game.



Figure 6. Advanced tasks.

4. Research result

4.1. Learning achievement of CT and AI concepts

Due to the data conforming to normal distribution and having homogeneity of variances, we utilized a one-way ANCOVA to analyze the results of the four groups. A post-hoc test was applied to examine the differences among the various groups. The post-hoc test compared each pair of groups (EG1 vs. EG2, EG1 vs. EG3, EG1 vs. CG, EG2 vs. EG3, EG2 vs. CG, and EG3 vs. CG); however, for clarity and conciseness, only statistically significant comparisons are reported in Table 3. The results of both Levene's test ($F = 1.982, p < .05$) and the assumption of regression homogeneity ($F = 0.383, p < .05$) indicate non-significance. According to the results presented in Table 3, there were significant differences in the posttest scores among the four groups ($F = 3.358, p < .05$). The post-hoc test indicated that the posttest scores of EG1, EG2, and EG3 were all significantly higher than those of the control group. The effect size ($\eta^2 = .079$) indicates a medium effect according to Cohen's (1988) benchmarks, suggesting that the interventions exerted a meaningful, though not large, impact on students' learning achievement.

4.2. CT self-efficacy

Since the data did not follow normal distribution, we utilized the Kruskal–Wallis H test to analyze the results of the four groups. A post-hoc U test was employed to assess the differences among the groups. The post-hoc U test compared all possible pairs of groups, but only the pairs showing statistically significant differences are presented in Table 4 to maintain concise reporting. As indicated in the results presented in Table 4, there were significant differences among the four groups in terms of Cooperative Learning and Creative Thinking. Based on the post-hoc U test results, in the Cooperative Learning dimension, EG1, EG3, and CG all scored significantly higher than EG2; in the Creative Thinking dimension, both EG1 and EG3 scored significantly higher than EG2.

4.3. Learning anxiety and engagement

As the data did not satisfy the assumptions of normal distribution, we utilized the Kruskal–Wallis H test to analyze the results of the four groups. A post-hoc U test was conducted to examine the differences

Table 3. Learning achievement of CT and AI concepts.

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted Mean	<i>SE</i>	<i>F</i>	<i>p</i>	η^2	post-hoc
EG1	39	65.41	12.40	66.03	1.93	3.358*	.021	.079	EG1 > CG*
EG2	28	66.71	9.60	66.95	2.27				EG2 > CG*
EG3	31	67.87	11.74	67.58	2.17				EG3 > CG**
CG	28	59.11	14.88	58.74	2.26				

* $p < .05$, ** $p < .01$.

Table 4. CT self-efficacy.

Dimensions	Group	N	Mean Rank	SD	H	post-hoc (<i>U</i> test)
Problem-solving	EG1	39	59.65	0.52	1.823	
	EG2	28	65.89	0.85		
	EG3	31	69.76	0.74		
	CG	28	59.54	0.70		
Cooperative Learning	EG1	39	76.33	0.80	28.154***	EG1 > EG2* EG3 > EG2*** CG > EG2*
	EG2	28	33.95	0.38		
	EG3	31	77.13	0.75		
	CG	28	60.09	0.58		
Creative Thinking	EG1	39	73.08	0.50	13.355**	EG1 > EG2* EG3 > EG2*
	EG2	28	47.04	0.64		
	EG3	31	74.63	0.58		
	CG	28	54.30	0.46		
Algorithmic Thinking	EG1	39	75.64	0.48	7.294	
	EG2	28	52.96	0.69		
	EG3	31	60.08	0.56		
	CG	28	60.91	0.54		

* $p < .05$, ** $p < .01$, *** $p < .001$.

among the groups. The post-hoc *U* test compared all possible pairs of groups, but only the pairs showing statistically significant differences are presented in Table 5 to maintain concise reporting. According to the findings presented in Table 5, significant differences were observed among the four groups in terms of Learning Anxiety. Based on the results of the post-hoc *U* test, EG3 scored significantly higher than both EG1 and EG2 in the dimension of Learning Anxiety.

4.4. Analysis of the sequential behavior patterns

To investigate changes in behavioral patterns, sequential behavior analyses were performed. Figure 7, Figure 8 and Figure 9 show the behavioral transition diagrams of the students involved in the three different methods. Table 6, Table 7 and Table 8, respectively, show the *Z* values for the three groups, revealing the relationships between the two codes. The behavioral sequence reaches a significant level ($p < 0.05$) when the *Z* value is more than 1.96 ($Z > 1.96$) (Bakeman & Gottman, 1997).

All the behavioral patterns (see Figure 2, Figure 3 and Figure 4) were statistically significant, where the *Z* value is represented on each line, and the direction of each line represents the direction of the behavioral transition.

Figure 5 depicts the behavioral pattern of the control group. Throughout the classroom activities, students frequently engaged in discussions and shared their ideas with peers (GC→EO) since they were required to collaboratively build and test speech recognition models with their teammates. During the process of dataset construction, students shared their ideas with others (EO) to ensure a smooth process. After discussion (GC), they proceeded to implement the actions (OT) related to dataset construction. During this process, students often discussed aspects such as enhancing dataset generality, determining the quantity of files for each training label, and distributing the workload. Whenever a student proposed an idea, it was followed by discussion with teammates before implementation (EO→GC→OT).

Table 5. Learning anxiety and engagement.

Dimensions	Group	N	Mean Rank	SD	H	post-hoc (<i>U</i> test)
Learning anxiety	EG1	39	42.42	0.39	28.643***	EG3 > EG1*** EG3 > EG2*
	EG2	28	62.86	0.75		
	EG3	31	89.27	0.62		
	CG	28	64.96	0.68		
Playfulness	EG1	39	54.90	0.38	7.095	
	EG2	28	60.29	0.61		
	EG3	31	77.60	0.59		
	CG	28	63.09	0.47		
Enjoyment	EG1	39	72.13	0.86	6.652	
	EG2	28	50.89	0.89		
	EG3	31	67.97	0.96		
	CG	28	59.14	0.62		

* $p < .05$, ** $p < .001$.

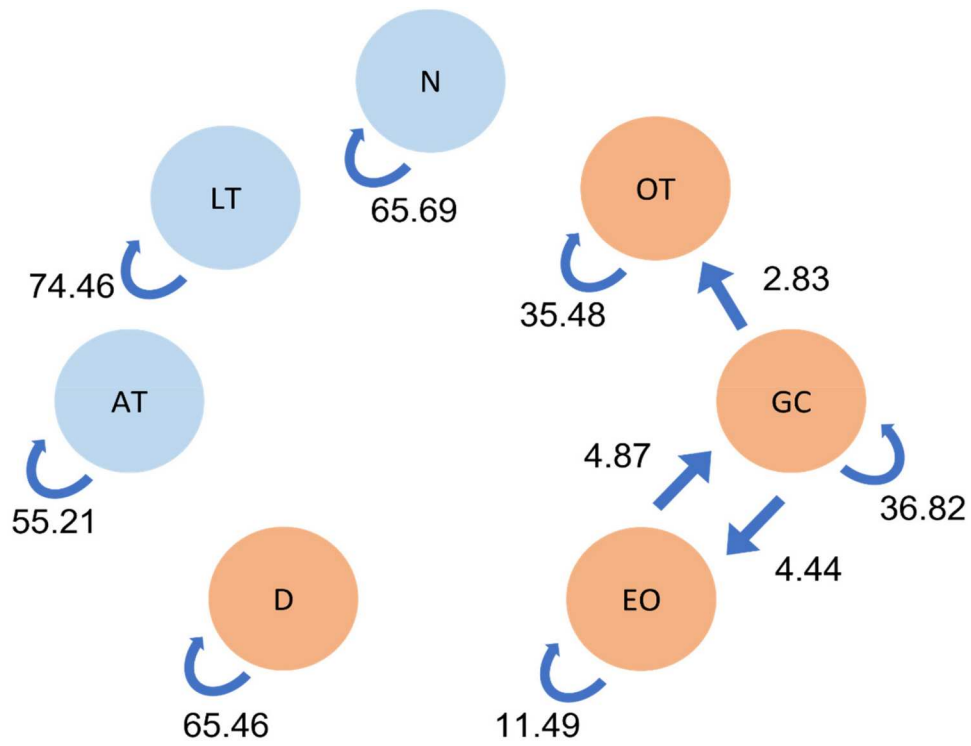


Figure 7. Sequential behavior patterns of the students in the control group.

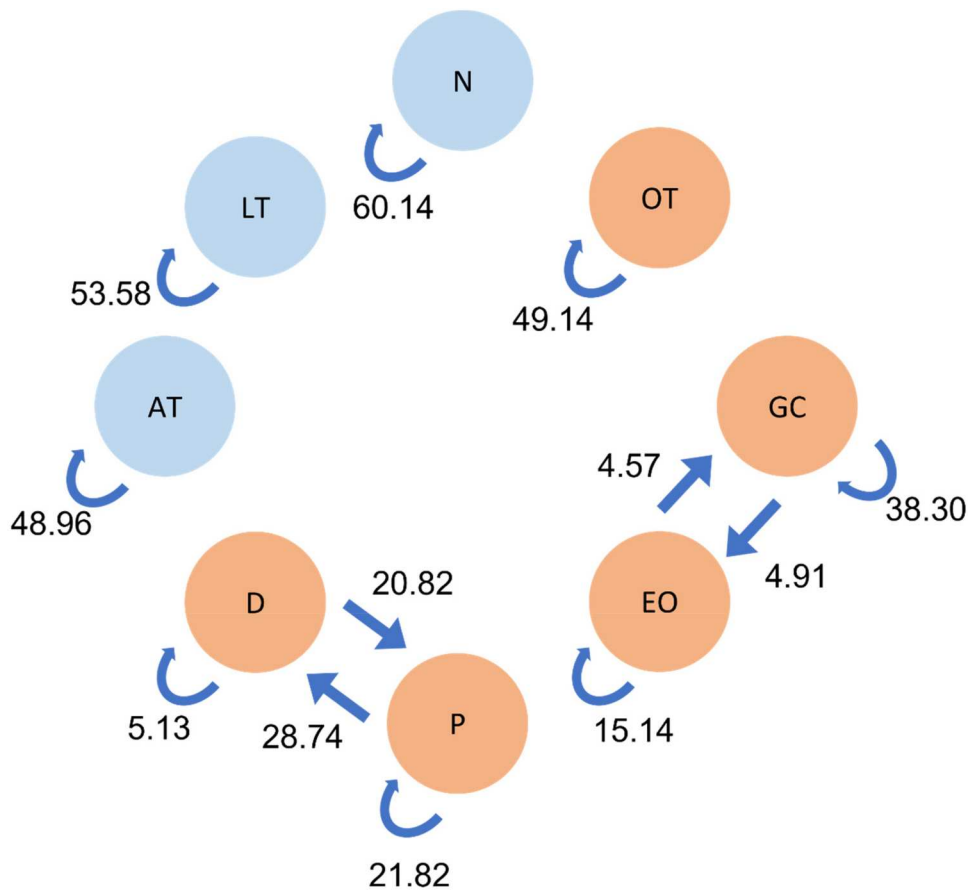


Figure 8. Sequential behavior patterns of the students in experimental group 3.

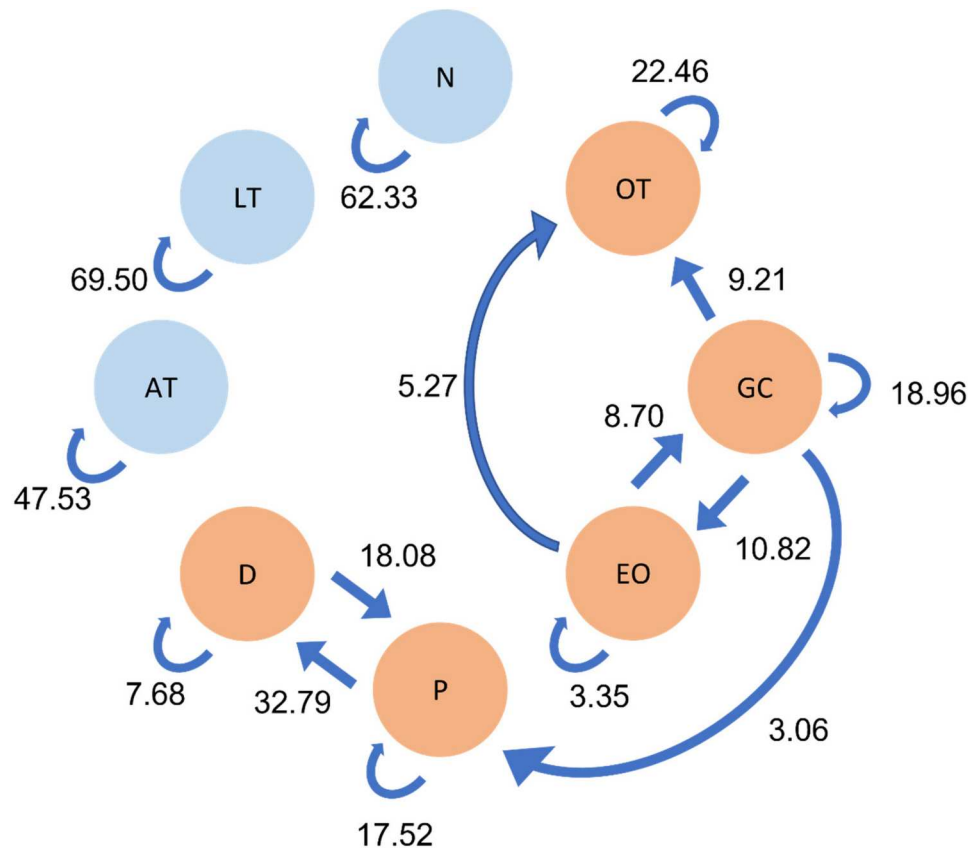


Figure 9. Sequential behavior patterns of the students in experimental group 1.

Table 6. Z Value of the control group.

Given	Target						
	LT	GC	AT	N	OT	EO	D
LT	74.46*	-23.96	-5.10	-16.60	-18.17	-3.98	-7.71
GC	-23.81	36.82*	-4.72	-19.29	2.83*	4.44*	-9.73
AT	-5.75	-4.35	55.21*	-5.02	-4.17	-0.42	-1.49
N	-16.07	-18.53	-4.59	65.69*	-18.32	-4.10	-7.10
OT	-18.40	1.44	-4.37	-16.68	35.48*	-1.29	-7.07
EO	-4.48	4.87*	-1.04	-4.10	-1.29	11.49*	0.87
D	-7.81	-8.71	-1.87	-7.36	-7.83	0.45	65.46*

* $p < .05$.

Table 7. Z value of experimental group 3.

Given:	Target							
	LT	GC	AT	P	N	OT	EO	D
LT	53.58*	-14.10	-4.27	-3.63	-14.15	-7.29	-4.60	-3.66
GC	-14.97	38.30*	-4.80	-1.49	-20.86	-6.54	4.91*	-7.43
AT	-4.95	-5.85	48.96*	1.24	-6.57	-2.66	-0.67	-1.16
P	-5.71	-3.52	1.34	21.82*	-13.23	-8.79	0.70	28.74*
N	-12.46	-20.86	-6.75	-14.09	60.14*	-12.69	-8.21	-7.43
OT	-6.76	-6.49	-3.77	-8.54	-12.41	49.14*	-1.17	-4.37
EO	-4.09	4.57*	-1.44	1.10	-7.37	-1.25	15.14*	-1.90
D	-3.36	-3.89	-2.14	20.82*	-7.23	-3.54	0.46	5.13*

* $p < .05$.

Figure 6 illustrates the behavioral pattern of EG3. When building and testing speech recognition models, students engaged in mutual discussion and exchanged opinions (GC→EO, EO→GC). Upon completing the setup to control the robot car in their app, they initiated testing. If errors arose, they returned to debugging,

Table 8. Z value of experimental group 1.

Given:	Target							
	LT	GC	AT	P	N	OT	EO	D
LT	69.50*	-24.63	-4.45	-10.03	-7.81	-12.85	-7.73	-8.15
GC	-24.49	18.96*	-5.88	3.06*	-12.73	9.21*	10.82*	-10.38
AT	-4.70	-4.62	47.53*	-3.48	-1.41	-4.72	-1.90	-3.11
P	-11.65	-1.43	-3.40	17.52*	-7.62	-12.26	-5.08	32.79*
N	-7.38	-13.57	0.31	-7.03	62.33*	-8.19	-5.05	-5.19
OT	-13.09	10.44	-4.75	-11.92	-7.91	22.46*	1.83	-5.11
EO	-7.35	8.70*	-2.80	-4.30	-5.02	5.27*	3.35*	-3.43
D	-7.32	-2.35	-1.86	18.08*	-4.75	-5.57	-1.75	7.68*

* $p < .05$.

followed by further testing ($D \rightarrow P$, $P \rightarrow D$). The students' robot car movements experienced issues if sensor calibration was not performed accurately, leading to deviations during gameplay. This prompted students to debug their setup after each round of play ($P \rightarrow D$).

Figure 7 presents the behavioral pattern of EG1. During the initial phase of model building and testing, students followed discussion and idea-sharing with immediate implementation ($EO \rightarrow GC \rightarrow OT$, $GC \rightarrow EO \rightarrow OT$). While conducting robot car testing and gameplay, discussion preceded the students' actions with the robot car ($GC \rightarrow P$). In case of errors, they backtracked to debugging, and after resolution, testing followed ($D \rightarrow P$, $P \rightarrow D$). Similar to EG3, the same situation occurred during gameplay, and this situation recurred more frequently ($P \rightarrow D$).

5. Discussion

5.1. Learning achievement

To examine the learning achievements of the four groups of students, we employed a one-way ANOVA and post-hoc tests. Based on the experimental results, all three experimental groups exhibited significantly higher learning achievements compared to the control group. These results indicate that regardless of whether AIoT board games, peer assessment, or a combination of both were employed, students' learning achievements were significantly higher than those who learned with traditional teaching methods.

This result underscores the effectiveness of peer assessment in enhancing students' learning achievements, as active participation in both learning and evaluative activities fosters greater engagement and responsibility in learning processes (Hodgson et al., 2014). Through peer assessment, students evaluate their peers' work, critically examine their own performance, and become more invested in the course, thereby elevating their overall learning achievements. This reflective process also promotes higher-order thinking, as students analyze and provide constructive feedback, deepening their understanding of the subject matter (Zhan et al., 2023).

Furthermore, the AIoT board games employed in this study provided students with a collaborative and interactive learning environment that facilitated face-to-face discussions, cognitive engagement, and teamwork, all of which enhance learning outcomes (Lin & Hou, 2022). The descriptive results revealed a performance trend of $EG3 > EG2 > EG1 > CG$. This pattern suggests that students who participated solely in the AIoT game condition (EG3) achieved slightly higher learning performance than those in the peer assessment (EG2) or combined intervention (EG1) groups. One plausible explanation is that integrating peer assessment into the AIoT gameplay may have increased task complexity and cognitive load, as students had to manage both strategic gameplay and evaluative responsibilities. According to cognitive load theory, engaging in multiple cognitively demanding tasks simultaneously can impose extraneous load, potentially diminishing short-term performance (Sweller, 2010).

Another possible explanation is related to learning motivation and flow. The AIoT board game alone might have elicited stronger intrinsic motivation and a higher level of flow experience due to its competitive and playful nature, resulting in higher immediate learning outcomes (Hamari et al., 2016). In contrast, adding peer assessment may have disrupted the game's immersive flow by introducing evaluative pauses that shifted students' focus toward reflection rather than immediate gameplay. Nevertheless, peer assessment is particularly valuable for developing reflective, metacognitive, and self-regulatory abilities, which may

not immediately manifest in performance scores but contribute to sustainable learning benefits over time (Double et al., 2020).

Ultimately, this study demonstrates that all three innovative approaches, AIoT game-based learning, peer assessment, and their combination, effectively enhance students' learning achievements compared with traditional instruction. The integrated approach (EG1) may therefore serve a complementary rather than additive role, balancing the motivational and collaborative benefits of gameplay with the reflective depth fostered by peer assessment.

5.2. CT self-efficacy

To examine the CT self-efficacy of the four student groups, we employed the Kruskal–Wallis H test and post-hoc U test. Based on the experimental results, significant differences were observed in the aspects of cooperative learning and creative thinking. However, no significant differences were found in problem-solving and algorithmic thinking. One possible explanation is that these dimensions typically require more extended practice and repeated opportunities for application to show measurable changes (Voogt et al., 2015). The short two-week intervention may not have been sufficient to yield observable improvements in these specific areas.

In terms of cooperative learning, the post-hoc U test results indicated that EG1, EG3, and CG exhibited significantly higher levels of CT self-efficacy compared to EG2. In the realm of creative thinking, the post-hoc U test results revealed that both EG1 and EG3 displayed significantly higher levels of CT self-efficacy than EG2.

In this study, students engaged in cooperative tasks while training voice recognition models. However, EG2 conducted their learning activities in an online environment, which made it more challenging for them to engage in effective cooperation. This difficulty likely contributed to their poorer performance in the aspect of cooperative learning. This modality difference highlights a limitation of the study, as the lower cooperative learning performance in EG2 may be partially attributed to the online learning context rather than the peer assessment intervention itself. On the other hand, the groups that participated in AIoT board games, as seen in this study, demonstrated better performance in the aspect of creative thinking. Engaging in AIoT board games likely stimulated new ideas among the students, aligning with previous research findings (Bayeck, 2023; Carter et al., 2014).

5.3. Learning anxiety and engagement

To examine the learning anxiety and engagement of the four student groups, we employed the Kruskal–Wallis H test and post-hoc U test. According to the experimental results, significant differences were found only in the aspect of learning anxiety. In contrast, playfulness and enjoyment did not show significant differences across groups. This may be because these affective dimensions are more closely related to students' intrinsic interest and prior attitudes toward games, which are less likely to change substantially within a short intervention period (Lin & Hou, 2022). Thus, while the AIoT game and peer assessment successfully influenced anxiety levels, they did not necessarily alter broader affective dispositions in the limited timeframe.

The post-hoc U test results indicated that EG3 had significantly higher levels of learning anxiety compared to both EG1 and EG2. To further examine whether this difference resulted from pre-existing individual variations, we analyzed the pretest data on learning anxiety. The results showed no significant differences among the four groups, suggesting that the higher anxiety observed in EG3 was not attributable to prior individual differences or baseline affective states. Instead, it may have arisen from situational factors during gameplay, such as time pressure or the competitive and problem-solving nature of the AIoT board game. This aligns with previous findings that challenge-induced tension in game-based learning can elicit facilitative anxiety, which enhances attention and persistence rather than hindering performance (Bandura, 1977; Zeidner, 1998). Prior studies have shown that in digital game-based learning environments, high-anxiety learners can still achieve strong learning performance when their engagement in gaming activities supports their learning processes (Yang et al., 2018). This suggests that the anxiety observed in EG3 might have coexisted with effective learning behaviors rather than directly impairing performance.

In contrast, the peer assessment groups exhibited lower learning anxiety. In this study, students in the peer assessment conditions were provided with structured evaluation forms outlining assessment criteria and performance indicators. These explicit rubrics guided students' project development and peer evaluation, reducing uncertainty and fostering confidence in their performance (Topping, 2009). Additionally, we further examined whether affective factors such as playfulness, enjoyment, and learning anxiety could predict or moderate learning performance. The regression analyses revealed no significant effects, suggesting that these affective constructs did not directly influence students' learning outcomes in this study.

It should also be noted that the online-only modality in EG2 may have acted as a potential confounding factor. The lack of face-to-face interaction could have limited opportunities for spontaneous communication and emotional support, thereby influencing both CT self-efficacy and learning anxiety. This limitation suggests that differences between EG2 and the other groups cannot be attributed solely to the peer assessment intervention but may also be partially explained by the online learning environment (Voogt et al., 2015).

Finally, the results of the sequential behavior analysis revealed that EG1's behavioral pattern was more complex compared with those of CG and EG3. This complexity reflects the benefits of peer assessment, which encouraged students to engage in meaningful communication, collaborative problem-solving, and iterative feedback processes. Students in CG and EG3, however, tended to spend more time working individually on debugging or task execution, indicating that peer assessment contributed to richer social and cognitive engagement.

6. Conclusion and limitations

This study examined the impact of AIoT board games and peer assessment on undergraduate students' learning of CT and AI. The results indicated that all three experimental conditions, AIoT board game only, peer assessment only, and the combined approach, significantly outperformed traditional instruction in learning achievement and CT self-efficacy. Descriptive results revealed a performance pattern of $EG3 > EG2 > EG1 > CG$, suggesting that while the AIoT board game alone yielded the highest short-term learning gains, the combined approach provided complementary benefits by integrating the motivational engagement of gameplay with the reflective and metacognitive processes fostered through peer assessment. Sequential behavior analysis further confirmed that peer assessment effectively facilitated discussions and interactions, enriching collaborative learning processes. Notably, despite undergoing online instruction, EG2 achieved higher learning gains than the control group, reinforcing the robustness of peer feedback as an effective learning strategy even in virtual environments. Future research should explore the applicability of these methods across disciplines to broaden their educational impact and sustainability.

Promoting AI education is essential in modern curricula, and integrating a personal audio classifier supports this goal. This tool provides an intuitive, hands-on approach to learning AI by engaging students in data collection, model training, and classifier implementation, reinforcing key CT skills such as abstraction, pattern recognition, and algorithmic thinking. Its scalability allows adaptation across educational levels, from simple command recognition to complex voice pattern applications, ensuring accessibility for diverse learners.

Beyond computer science, the personal audio classifier facilitates interdisciplinary learning, linking AI with fields such as language arts and music. It fosters active learning, engagement, and innovation by enabling students to create and refine their models, enhancing motivation and knowledge retention. Additionally, its use in open-source platforms ensures accessibility for underprivileged communities, promoting equitable AI education.

Moreover, the tool encourages ethical awareness by exposing students to real-world AI challenges, including data privacy and bias. Educators play a critical role in guiding discussions on responsible AI use. In conclusion, the personal audio classifier is a versatile, impactful tool that simplifies AI concepts, engages learners, and broadens AI literacy. Its integration into education can equip students with essential skills for an AI-driven future.

Despite the positive findings, this study has several limitations. First, EG2 was conducted online due to the COVID-19 pandemic, whereas the other groups were conducted face-to-face. This difference in modality may

have confounded the effects of peer assessment, particularly with respect to cooperative learning outcomes. Second, the learning achievement test showed relatively low reliability (KR-20 values below .70), which may have introduced measurement errors. Although the CT self-efficacy scale demonstrated acceptable internal consistency after item selection, further validation studies are recommended to strengthen its applicability in different contexts. Finally, the short intervention period of two weeks (six hours in total) is acknowledged as a limitation, since cultivating higher-order thinking skills such as critical and computational thinking requires repeated practice, iterative reflection, and longer-term engagement (Voogt et al., 2015; Wing, 2011). Short-term interventions may primarily influence surface-level behaviors, such as modifying assignments according to criteria, rather than fundamentally reshaping thinking modes. Future studies should therefore adopt extended interventions, incorporate repeated practice cycles, and include delayed post-tests to capture the gradual and sustained development of higher-order thinking, in line with the progressive nature of educational learning processes.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study is supported in part by the National Science and Technology Council under contract numbers NSTC 111-2410-H-003-168-MY3 and NSTC 113-2628-H-003-002.

Author contributions

Mu-Sheng Chen: Conceptualization, Methodology, Data curation, Software, Formal Analysis, Writing – Original draft preparation. **Ting-Chia Hsu:** Supervision, Funding acquisition, Conceptualization, Methodology, Validation, Writing – Reviewing and Editing.

ORCID

Mu-Sheng Chen  <http://orcid.org/0000-0002-0627-2424>
 Ting-Chia Hsu  <http://orcid.org/0000-0001-6504-9540>

References

- Alt, D. (2023). Assessing the benefits of gamification in mathematics for student gameful experience and gaming motivation. *Computers & Education*, 200, 104806. <https://doi.org/10.1016/j.compedu.2023.104806>
- Bakeman, R., & Gottman, J. M. (1997). *Observing interaction: An introduction to sequential analysis*. Cambridge university press.
- Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review*, 84(2), 191–215. <https://doi.org/10.1037/0033-295X.84.2.191>
- Barr, V., & Stephenson, C. (2011). Bringing computational thinking to K-12: What is Involved and what is the role of the computer science education community? *Acm Inroads*, 2(1), 48–54. <https://doi.org/10.1145/1929887.1929905>
- Bayeck, R. Y. (2023). Understanding computational thinking in the gameplay of the African Songo board game. *British Journal of Educational Technology*, 55(1), 259–276. <https://doi.org/10.1111/bjet.13353>
- Carter, M., Gibbs, M., & Harrop, M. (2014). Drafting an army. *Games and Culture*, 9(2), 122–147. <https://doi.org/10.1177/1555412013513349>
- Cartney, P. (2010). Exploring the use of peer assessment as a vehicle for closing the gap between feedback given and feedback used. *Assessment & Evaluation in Higher Education*, 35(5), 551–564. <https://doi.org/10.1080/02602931003632381>
- Cartney, P., & Rouse, A. (2006). The emotional impact of learning in small groups: Highlighting the impact on student progression and retention. *Teaching in Higher Education*, 11(1), 79–91. <https://doi.org/10.1080/13562510500400180>
- Cassidy, S. (2006). Developing employability skills: Peer assessment in higher education. *Education + Training*, 48(7), 508–517. <https://doi.org/10.1108/00400910610705890>

- Chang, C.-K. (2014). Effects of using Alice and Scratch in an introductory programming course for corrective instruction. *Journal of Educational Computing Research*, 51(2), 185–204. <https://doi.org/10.2190/EC.51.2.c>
- Chang, C. Y., Lee, D. C., Tang, K. Y., & Hwang, G. J. (2021). Effect sizes and research directions of peer assessments: From an integrated perspective of meta-analysis and co-citation network. *Computers & Education*, 164, 104123. <https://doi.org/10.1016/j.compedu.2020.104123>
- Charlier, N., & De Fraine, B. (2013). Game-based learning as a vehicle to teach first aid content: A randomized experiment. *Journal of School Health*, 83(7), 493–499.
- Cheng, K. H., & Tsai, C. C. (2012). Students' interpersonal perspectives on, conceptions of and approaches to learning in online peer assessment. *Australasian Journal of Educational Technology*, 28(4), 599–618. <https://doi.org/10.14742/ajet.830>
- Cheng, Y. C. (2018). The effect of using board games in reducing language anxiety and improving oral performance. *Electronic Theses and Dissertations*, 899. <https://egrove.olemiss.edu/etd/899>
- Clark, D. B., Tanner-Smith, E. E., & Killingsworth, S. S. (2016). Digital Games, Design, and Learning. *Review of Educational Research*, 86(1), 79–122. <https://doi.org/10.3102/0034654315582065>
- Clegg, K., & Bryan, C. (2006). *Reflections, rationales and realities*. Routledge.
- Cohen, J. (1988). *Statistical power analysis for the behavioral Sciences*. Lawrence Erlbaum Associates.
- Cole, D. A. (1991). Change in self-perceived competence as a function of peer and teacher evaluation. *Developmental Psychology*, 27(4), 682–688. <https://doi.org/10.1037/0012-1649.27.4.682>
- DeNero, J., & Klein, D. (2010, July). Teaching introductory artificial intelligence with pac-man. In *First AAAI Symposium on Educational Advances in Artificial Intelligence*.
- Double, K. S., McGrane, J. A., & Hopfenbeck, T. N. (2020). The impact of peer assessment on academic performance: A meta-analysis of control group studies. *Educational Psychology Review*, 32(2), 481–509. <https://doi.org/10.1007/s10648-019-09510-3>
- Ehly, S. W., & Topping, K. J. (1998). *Peer-assisted learning*. L. Erlbaum Associates.
- Gee, J. P. (2005). Learning by design: Good video games as learning machines. *E-learning and Digital Media*, 2(1), 5–16. <https://doi.org/10.2304/elea.2005.2.1.5>
- Hamari, J., Shernoff, D. J., Rowe, E., Coller, B., Asbell-Clarke, J., & Edwards, T. (2016). Challenging games help students learn: An empirical study on engagement, flow, and immersion in game-based learning. *Computers in Human Behavior*, 54, 170–179. <https://doi.org/10.1016/j.chb.2015.07.045>
- Hinebaugh, J. P. (2009). *A board game education*. R&L Education.
- Hodgson, P., Chan, K., & Liu, J. (2014). Outcomes of synergetic peer assessment: First-year experience. *Assessment & Evaluation in Higher Education*, 39(2), 168–178. <https://doi.org/10.1080/02602938.2013.803027>
- Li, L. (2017). The role of anonymity in peer assessment. *Assessment & Evaluation in Higher Education*, 42(4), 645–656. <https://doi.org/10.1080/02602938.2016.1174766>
- Lin, Y. C., & Hou, H. T. (2022). The evaluation of a scaffolding-based augmented reality educational board game with competition-oriented and collaboration-oriented mechanisms: Differences analysis of learning effectiveness, motivation, flow, and anxiety. *Interactive Learning Environments*, 502–521. <https://doi.org/10.1080/10494820.2022.2091606>
- Oktaviane, J., Prestiliano, J., & Prasida, T. A. S. (2021). Designing educational board games for children on snake handling using competitive mechanisms. *Indonesian Journal of Educational Research and Review*, 4(1), 1–8. <https://doi.org/10.23887/ijerr.v4i1.30443>
- Panadero, E., & Alqassab, M. (2019). An empirical review of anonymity effects in peer assessment, peer feedback, peer review, peer evaluation and peer grading. *Assessment & Evaluation in Higher Education*, 44(8), 1253–1278.
- Panadero, E., Jonsson, A., & Strijbos, J. W. (2016). Scaffolding self-regulated learning through self-assessment and peer assessment: Guidelines for classroom implementation. In D. Laveault, & L. Allal (Eds.), *Assessment for Learning: Meeting the Challenge of Implementation* (pp. 311–326). Springer.
- Schmidhuber, J. (2015). Deep learning in neural networks: An overview. *Neural networks*, 61, 85–117. <https://doi.org/10.1016/j.neunet.2014.09.003>
- Selby, C. & Woollard, J. (2013). *Computational thinking: The developing definition* (pp. 6). University of Southampton (E-prints).
- Sweller, J. (2010). Element interactivity and intrinsic, extraneous, and germane cognitive load. *Educational Psychology Review*, 22(2), 123–138. <https://doi.org/10.1007/s10648-010-9128-5>
- Topping, K. J. (2009). Peer assessment. *Theory into Practice*, 48(1), 20–27. <https://doi.org/10.1080/00405840802577569>
- Topping, K. J. (2023). Digital peer assessment in school teacher education and development: A systematic review. *Research Papers in Education*, 38(3), 472–498. <https://doi.org/10.1080/02671522.2021.1961301>
- Topping, K. J., Smith, E. F., Swanson, I., & Elliot, A. (2000). Formative peer assessment of academic writing between post-graduate students. *Assessment & Evaluation in Higher Education*, 25(2), 149–169. <https://doi.org/10.1080/713611428>
- Venkatesh, V. (2000). Determinants of perceived ease of use: Integrating control, intrinsic motivation, and emotion into the technology acceptance model. *Information Systems Research*, 11(4), 342–365.
- Voogt, J., Fisser, P., Good, J., Mishra, P., & Yadav, A. (2015). Computational thinking in compulsory education: Towards an agenda for research and practice. *Education and Information Technologies*, 20, 715–728. <https://doi.org/10.1007/s10639-015-9412-6>
- Wing, J. (2011). Research notebook: Computational thinking – What and why. *The Link Magazine*, 6, 20–23.

- Wing, J. M. (2006). Computational thinking. *Communications of the ACM*, 49(3), 33–35.
- Wu, C.-J., Chen, G.-D., & Huang, C.-W. (2014). Using digital board games for genuine communication in EFL classrooms. *Educational Technology Research and Development*, 62(2), 209–226. <https://doi.org/10.1007/s11423-013-9329-y>
- Yadav, A., Mayfield, C., Zhou, N., Hambrusch, S., & Korb, J. T. (2014). Computational thinking in elementary and secondary teacher education. *ACM Transactions on Computing Education*, 14(1), 1–16. <https://doi.org/10.1145/2576872>
- Yağcı, M. (2019). A valid and reliable tool for examining computational thinking skills. *Education and Information Technologies*, 24(1), 929–951. <https://doi.org/10.1007/s10639-018-9801-8>
- Yang, J. C., Lin, M. Y. D., & Chen, S. Y. (2018). Effects of anxiety levels on learning performance and gaming performance in digital game-based learning. *Journal of Computer Assisted Learning*, 34(3), 324–334. <https://doi.org/10.1111/jcal.12245>
- Yorke, M. (2003). Formative assessment in higher education: Moves towards theory and the enhancement of pedagogic practice. *Higher Education*, 45(4), 477–501. <https://doi.org/10.1023/A:1023967026413>
- Zeidner, M. (1998). *Test anxiety: The state of the art*. Springer.
- Zhan, Y., Wan, Z. H., & Sun, D. (2022). Online formative peer feedback in Chinese contexts at the tertiary Level: A critical review on its design, impacts and influencing factors. *Computers & Education*, 176, 104341. <https://doi.org/10.1016/j.compedu.2021.104341>
- Zhan, Y., Yan, Z., Wan, Z. H., Wang, X., Zeng, Y., Yang, M., & Yang, L. (2023). Effects of online peer assessment on higher-order thinking: A meta-analysis. *British Journal of Educational Technology*, 54(4), 817–835. <https://doi.org/10.1111/bjet.13310>